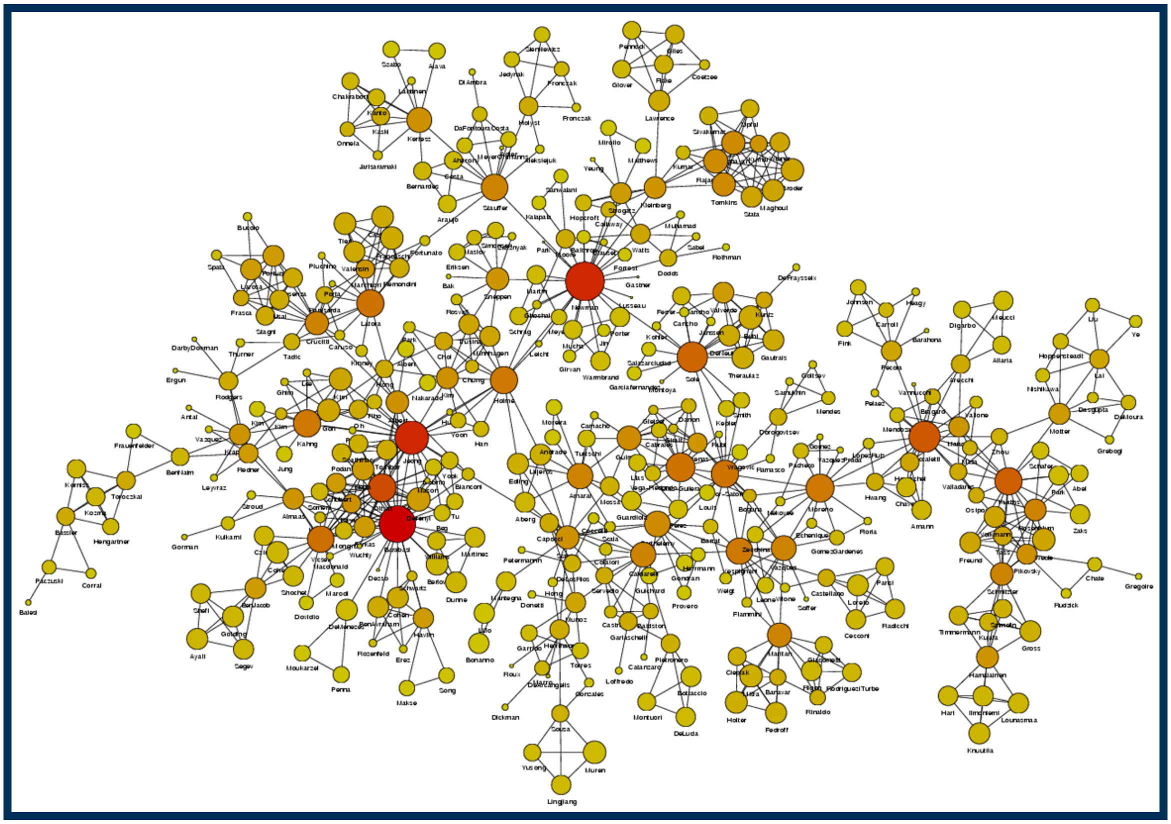
L1: Network Centrality



Above is an image that shows the co-authorship network for a set of Network Science researchers: each node represents a researcher and two nodes are connected if they have published at least one paper together. *(Image Source: **[University of Michigan](http://www-personal.umich.edu/~mejn/centrality/labeled.png)** **(http://www-personal.umich.edu/~mejn/centrality/labeled.png)**)*

A very common question in network science is: given a network representation of a system, which are the most important modules or nodes? Or, which are the most important connections or edges?

Of course, this depends on what we mean by "important" – and there are several different metrics that quantify the "centrality" of nodes and edges.

Sometimes we want to identify nodes and edges that are very central in the sense that most pairs of other nodes communicate through them.

Or, nodes and edges that, if removed, will cause the largest disruption in the underlying network.

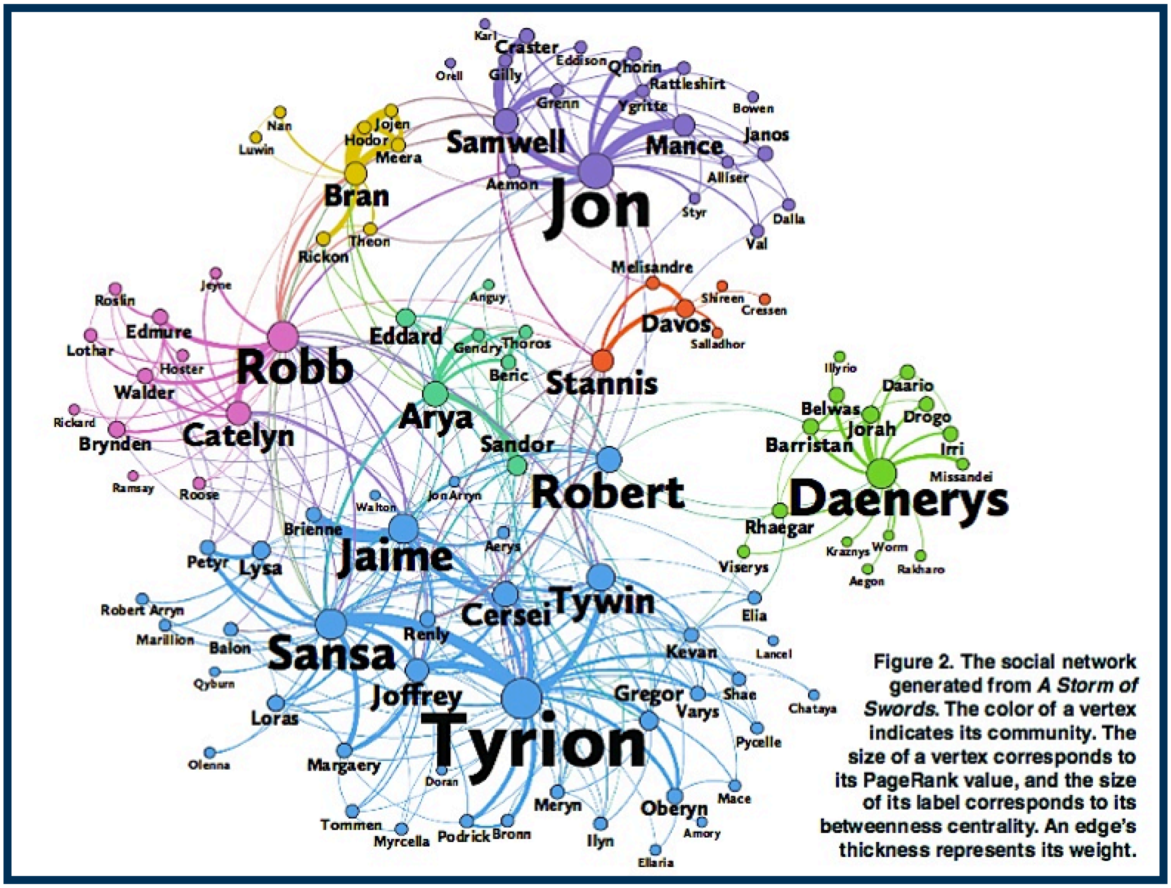


Figure 2. The social network generated from *A Storm of Swords*. The color of a vertex indicates its community. The size of a vertex corresponds to its PageRank value, and the size of its label corresponds to its betweenness centrality. An edge's thickness represents its weight.

For example, the image above refers to the characters of the third Game of Thrones novel, called ***"A storm of swords".*** *(Image Source:* The Measurement Standard, Carma)

Two nodes are connected if the corresponding two characters interacted in that novel, and the weight of the edge represents the length of that interaction.

Two different node centrality metrics are visualized in this figure. The size of the node refers to a centrality metric called PageRank – it is the same metric that was used by Google in their first web search engine. The PageRank value of a node v does not depend simply on how many other nodes point to v, but also how large their PageRank value is and how many other nodes they point to.

The second centrality metric refers to a centrality metric called "Betweenness" and it is shown by the size of the node's label. The Betweenness centrality of a node v relates to the number of shortest paths that traverse node v, considering the shortest paths across all node pairs.

Both metrics suggest that Tyrion and Jon are the most central characters in that novel, even though they were not interacting yet.

Source Links

[Finding community structure in networks using the eigenvectors of matrices (https://arxiv.org/abs/physics/0605087)](https://arxiv.org/abs/physics/0605087)